Q P Code:-25900512

Mahatma Gandhi University, Kottayam

MGU-UGP (HONOURS) Regular Examination October 2025

Third SEMESTER

Discipline Specific Elective (DSE)- MG3DSEMAT201
Game Theory and Project Management

(2024 Admission Onwards)

Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A

(Short Answer Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 What is meant by n-person game? **[K] / [C01]**

- Games which involve more than two players are called n-person game

. . . 2 Marks

Question 2 Determine whether the game has a saddle point whose payoff matrix is given by **[U] / [CO1]**

Player A	**Player B**	
	B_1	B_2
A_1	1	4
A_2	5	2

- $\min\limits_{j} \max\limits_{i} a_{ij} = 4$ **. . . 1 Mark**
- $\max\limits_{i} \min\limits_{j} a_{ij} = 2$ **. . . 1 Mark**
- **Conclusion:** The game has no saddle point **. . . 1 Mark**

Maximum Marks should not exceed 2

Question 3 Find optimal strategies and net outcome using the arithmetic method. **[A] / [CO1]**

Player A	Player B	
	B_1	B_2
A_1	5	1
A_2	2	4

- For finding the value of the game (3) by arithmetic method ... **2 Marks**
- For finding Player A's optimal strategy: $\left(\frac{1}{3}, \frac{2}{3}\right)$ ... **2 Marks**
- For finding Player B's optimal strategy: $\left(\frac{1}{2}, \frac{1}{2}\right)$ ... **2 Marks**

Maximum Marks should not exceed 2

Question 4 Explain the situations when graphical method is used **[U] / [CO2]**

- Player A has 2 strategies and Player B has n strategies (2 × n game) ... **1 Mark**
- Player B has 2 strategies and Player A has m strategies (m × 2 game) ... **1 Mark**

Question 5 Draw AON Network for the following list of activities **[U] / [CO3]**

Activity	Predecessor Activity
A	–
B	–
C	–
D	–
E	A, B, C
F	B, C
G	A, C
H	D, E

- For correct drawing ... **2 Marks**
- Drawing with minor mistakes ... **1 Mark**

Question 6 Define cycling in a network **[U] / [CO3]**

- Correct definition ... **2 Marks**

Question 7 How do you calculate latest occurrence time of the initial event? **[K] / [CO4]**

- This is the minimum of the latest start times of all activities from the initial event. ... **2 Marks**

Question 8 Define latest allowable time of an event **[K] / [CO4]**

- This is the latest time at which an event can occur without causing a delay in project's completion time. ... **2 Marks**

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 With an appropriate example, explain maximin and minimax principles **[U] / [CO1]**

- For writing maximin and minimax principles ... **3 Marks**
- For correct example ... **3 Marks**
- Explanation ... **2 Marks**

Maximum Marks should not exceed 6

Question 10 Find the saddle point of the following two-person zero-sum game **[U] / [CO1]**

Player A	Player B		
	B_1	B_2	B_3
A_1	6	5	7
A_2	5	4	6
A_3	8	5	9

- $\min\limits_{j} \max\limits_{i} a_{ij} = 5$ ... **3 Marks**
- $\max\limits_{i} \min\limits_{j} a_{ij} = 5$ ... **2 Marks**
- **Conclusion:** The saddle point is 5 ... **1 Mark**

Question 11 Determine the optimal strategies for both players and compute the value of the game for player A using the algebraic method. **[A] / [CO2]**

Player A	Player B	
	B_1	B_2
A_1	6	1
A_2	-3	2

- For finding the value of the game $(\frac{3}{2})$ by algebraic method ... **2 Marks**
- For finding Player A's optimal strategy: $\left(\frac{1}{2}, \frac{1}{2}\right)$ ... **2 Marks**
- For finding Player B's optimal strategy: $\left(\frac{1}{10}, \frac{9}{10}\right)$ ... **2 Marks**

Question 12 Formulate an equivalent linear programming model for player A and player B based on the zero-sum game defined by the given payoff matrix. **[A] / [CO2]**

Player A	Player B		
	B_1	B_2	B_3
A_1	6	8	2
A_2	8	2	10
A_3	4	10	12

- Player A wants to maximize the value v of the game: Using the transformation $x_i' = \frac{x_i}{v}$ (since $v > 0$), we get the standard LP form:

$$\text{Minimize } x_1' + x_2' + x_3'$$

subject to:

$$6x_1' + 8x_2' + 4x_3' \geq 1$$
$$8x_1' + 2x_2' + 10x_3' \geq 1$$
$$2x_1' + 10x_2' + 12x_3' \geq 1$$
$$x_1', x_2', x_3' \geq 0$$

. . . 6 Marks

OR

- Player B wants to minimize the value u of the game. Using the transformation $y_j' = \frac{y_j}{u}$, we get the standard LP form:

$$\text{Maximize } y_1' + y_2' + y_3'$$

subject to:

$$6y_1' + 8y_2' + 2y_3' \leq 1$$
$$8y_1' + 2y_2' + 10y_3' \leq 1$$
$$4y_1' + 10y_2' + 12y_3' \leq 1$$
$$y_1', y_2', y_3' \geq 0$$

. . . 6 Marks

Question 13 Explain Activity-on-Node network **[K] / [CO3]**

- For correct Explanation **. . . 6 Marks**

Question 14 Explain PERT and its significance in Project Management **[U] / [CO3]**

- For explaining PERT **. . . 3 Marks**
- For writing its significance **. . . 3 Marks**

Question 15 The following table gives the list of various activities and their immediate predecessors involved in a Computer Software Project. **[A] / [CO4]**

Activity	Predecessor activity	Duration(Days)
A	$--$	5
B	$--$	7
C	$--$	9
D	A, B, C	8
E	A, B, C	5
F	A, B, C	2
G	F	7
H	C	3
I	G	4
J	H, E	3
K	F	5
L	I, J	10

- For correct network diagram ... **4 Marks**
- For finding critical path by forward and backward pass method ... **3 Marks**
- For identifying critical and noncritical activities ... **2 Marks**

Maximum Marks should not exceed 6

Question 16 Discuss the possible values of Total Float in project scheduling. For each case, explain the reasons for its occurrence and suggest suitable remedies. **[U] / [CO4]**

- For writing possible values of Total Float in project scheduling ... **4 Marks**
- For correct explanation of its reason and remedies. ... **3 Marks**

Maximum Marks should not exceed 6

Part C
(Essay Questions)
Answer any 3 questions
Each question carries 10 marks

Question 17 Obtain the strategies for each player and the value of the game for the following two-person zero-sum game by using principle of dominance **[U] / [CO1]**

Player A	Player B			
	B_1	B_2	B_3	B_4
A_1	8	2	7	6
A_2	3	6	5	6
A_3	5	5	4	5

- Column $B_2 = (2, 6, 5)$ dominates column $B_4 = (6, 6, 5)$ since:

$$2 \leq 6, \quad 6 \leq 6, \quad 5 \leq 5,$$

. Hence, B_4 is removed.

Reduced matrix:

	B_1	B_2	B_3
A_1	8	2	7
A_2	3	6	5
A_3	5	5	4

. . . **10 Marks**

Question 18 Using the method of row minima and column maxima, solve the following payoff matrix. Check whether the game has a saddle point. If yes, find the value of the game and strategies of the players: **[E] / [CO2]**

Player A	Player B		
	B_1	B_2	B_3
A_1	7	3	5
A_2	4	6	2
A_3	8	1	9

- $\min\{7, 3, 5\} = 3 \quad (\text{for } A_1)$,
  $\min\{4, 6, 2\} = 2 \quad (\text{for } A_2)$,
  $\min\{8, 1, 9\} = 1 \quad (\text{for } A_3)$.

  Thus the **maximin** value is

$$\max\{3, 2, 1\} = 3.$$

. . . **4 Marks**

- $\max\{7, 4, 8\} = 8 \quad (\text{for } B_1)$,
  $\max\{3, 6, 1\} = 6 \quad (\text{for } B_2)$,
  $\max\{5, 2, 9\} = 9 \quad (\text{for } B_3)$.

  Thus the **minimax** value is

$$\min\{8, 6, 9\} = 6.$$

. . . **4 Marks**

- **Conclusion:** Since the maximin = 3 is not equal to the minimax = 6, the game **does not** have a saddle point. . . . **2 Marks**

Question 19 For the game given by the payoff matrix (Player A): **[E] / [CO2]**

	B_1	B_2	B_3
A_1	4	-2	3
A_2	2	1	0
A_3	5	2	1

(a) Use the dominance principle to simplify the game.

(b) Solve the reduced game using the matrix method.

(c) Find the value of the game.

- Row $A_3 = (5, 1, 2)$ dominates column $A_2 = (2, 1, 0)$ and the matrix reduces to

	B_1	B_2	B_3
A_1	4	-2	3
A_3	5	2	1

. . . 3 Marks

- Column $B_3 = (3, 1)$ dominates column $B_1 = (4, 5)$ and the matrix reduces to

	B_2	B_3
A_1	-2	3
A_3	2	1

. . . 3 Marks

- Calculations to find Value of the game $(\frac{4}{3})$ by matrix method. **. . . 3 Marks**
- Player A's optimal strategy in othe game is: $\left(\frac{1}{6},\ 0,\ \frac{5}{6}\right)$ **. . . 2 Marks**
- Player B's optimal strategy in othe game is: $\left(0,\ \frac{1}{3},\ \frac{2}{3}\right)$ **. . . 2 Marks**

Maximum Marks should not exceed 10

Question 20 Following is the list of activities associated with the assembly of a space module for an upcoming mission. Draw the AOA network diagram of activities involved in the

project. **[U] / [CO3]**

Activity	Description	Predecessor Activity
A	Construct shell of module	–
B	Order life support system	–
C	Order components of control	–
D	Wire module	A
E	Assemble control	C
F	Preliminary test	B
G	Install module	D, F
H	Install scientific experimentation	D, F
I	Preliminary test of control	E, F
J	Install navigational system	H, I
K	Final testing and debugging	G, J

- For Correct Network diagram ... **10 Marks**
- Network diagram with minor mistakes ... **6 Marks**

Question 21 Following table gives the list of various activities and their immediate predecessors involved in the CAT scanner in a hospital. **[A] / [CO4]**

Activity	Description	Predecessors	Duration(days)
A	Finalization of the layout plans	–	6
B	Demolition of the structure	–	9
C	Walls erection	A, B	15
D	Flooring	B	8
E	Electrical wiring	C	7
F	Air conditioning ducting	C	4
G	Fire alarm installation	C	3
H	False ceiling and light fittings	E, F, G	10
I	Wall plastering and painting	H	9
J	Equipment installation	D, I	6
K	Calibration and testing	J	3
L	Final finishing	K	2
M	Handing over	L, K	5

(a) Prepare a network diagram showing the interrelationship of the various activities.

(b) Find the total time required to hold the conference.

(c) Compute the total float for the noncritical activities.

- For Correct Network diagram ... **7 Marks**

- For finding the total time required to hold the conference by forward and backward pass method

  ...**3 Marks**

- For computing total float for the noncritical activities. ...**3 Marks**

Maximum Marks should not exceed 10

Question 22 The owner of a chain of fast food restaurants is considering a new computer system for accounting and inventory control. A computer company sent the following information about the computer system installation. **[A]/[CO4]**

Activity	Activity Description	Predecessors	Time Duration
A	Select the computer model	–	4
B	Design input/output system	A	15
C	Design monitoring system	A	12
D	Assemble computer hardware	B	25
E	Develop the main programmes	B	26
F	Develop input/output routines	C	16
G	Create database	E	12
H	Install the system	D, F	1
I	Test and implement	G, H	6

(a) Construct the network diagram for the project.

(b) Determine the critical path and compute the expected completion time.

(c) Find Total Float for each activities

- For Correct Network diagram ...**7 Marks**
- For finding critical path and the expected completion time by forward and backward pass method ...**3 Marks**
- For computing total float all activities. ...**3 Marks**

Maximum Marks should not exceed 10

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.